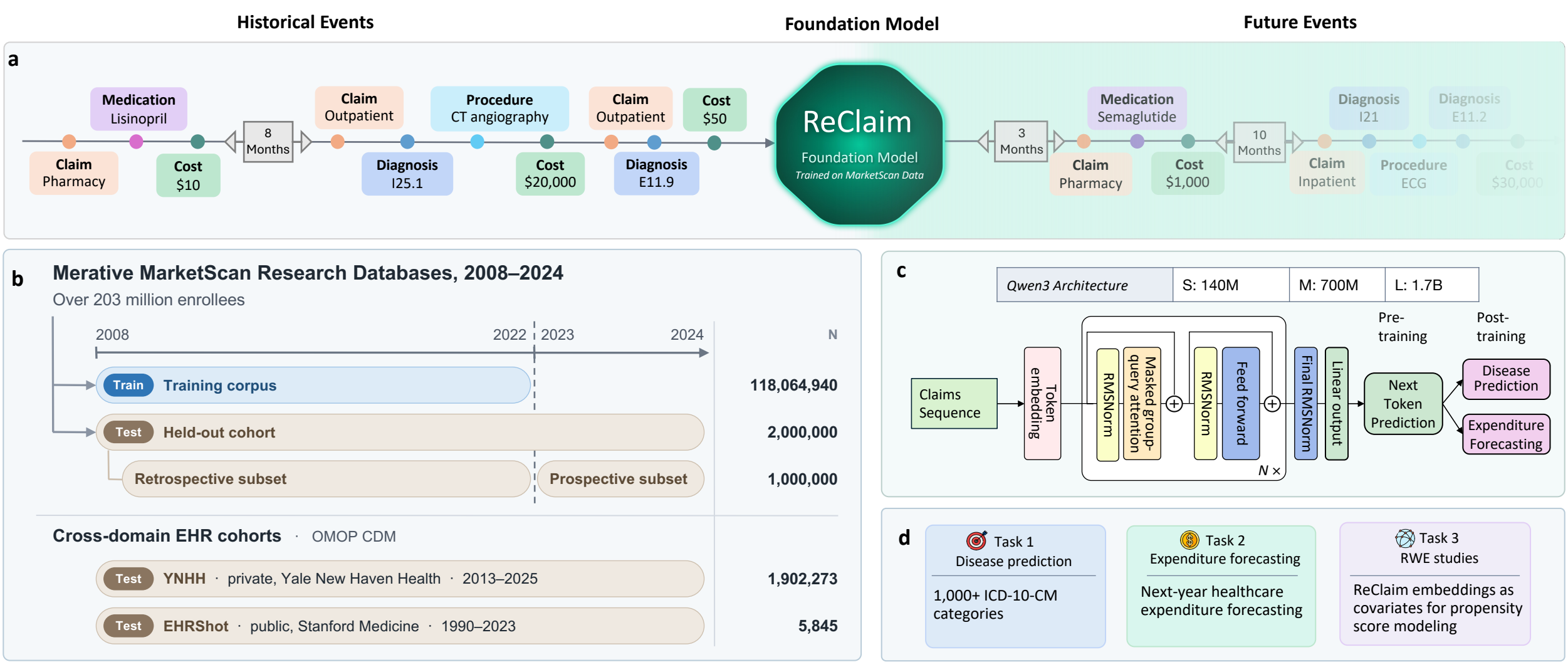

Historical Events
Foundation Model
Future Events
a
Medication
Lisinopril
Claim
Pharmacy
Cost
$10
8
Months
Claim
Outpatient
Diagnosis
I25.1
Procedure
CT angiography
Cost
$20,000
Claim
Outpatient
Diagnosis
E11.9
Cost
$50
ReClaim
Foundation Model
Trained on MarketScan Data
3
Months
Medication
Semaglutide
Claim
Pharmacy
Cost
$1,000
10
Months
Diagnosis
I21
Claim
Inpatient
Diagnosis
E11.2
Procedure
ECG
Cost
$30,000
b
Merative MarketScan Research Databases, 2008–2024
Over 203 million enrollees
2008
2022
2023
2024
N
Train
Training corpus
118,064,940
Test
Held-out cohort
2,000,000
Retrospective subset
Prospective subset
1,000,000
Cross-domain EHR cohorts · OMOP CDM
Test
YNHH · private, Yale New Haven Health · 2013–2025
1,902,273
Test
EHRShot · public, Stanford Medicine · 1990–2023
5,845
c
Qwen3 Architecture
S: 140M
M: 700M
L: 1.7B
Pre-training
Post-training
Claims Sequence
Token embedding
RMSNorm
Masked group-query attention
RMSNorm
Feed forward
N ×
Final RMSNorm
Linear output
Next Token Prediction
Disease Prediction
Expenditure Forecasting
d
Task 1
Disease prediction
1,000+ ICD-10-CM categories
Task 2
Expenditure forecasting
Next-year healthcare expenditure forecasting
Task 3
RWE studies
ReClaim embeddings as covariates for propensity score modeling